

Review

Human-algorithm interactions help explain the spread of misinformation

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Human attention biases toward moral and emotional information are as prevalent online as they are offline. When these biases interact with content algorithms that curate social media users' news feeds to maximize attentional capture, moral and emotional information are privileged in the online information ecosystem. We review evidence for these human-algorithm interactions and argue that misinformation exploits this process to spread online. This framework suggests that interventions aimed at combating misinformation require a dual-pronged approach that combines person-centered and design-centered interventions to be most effective. We suggest several avenues for research in the psychological study of misinformation sharing under a framework of human-algorithm interaction.

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Current Opinion in Psychology 2024, **56**:101770

This review comes from a themed issue on **The Psychology of Misinformation (2024)**

Edited by **Gordon Pennycook** and **Lisa K. Fazio**

For complete overview about the section, refer [The Psychology of Misinformation \(2024\)](#)

Available online 23 November 2023

<https://doi.org/10.1016/j.copsyc.2023.101770>

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Attention biases and algorithmic amplification

Following the 2016 U.S. presidential election, online political misinformation surged, with many political and social costs [1–4]. Debates continue about the prevalence of misinformation online [5–8], but understanding why people share it remains crucial for improving the online information ecosystem [9]. To date, psychological work has focused on factors such as inattention to accuracy and partisan bias in order to explain why people

share misinformation [10,11]. However, this work does not focus on how content algorithms - recommender systems that curate the information presented to users - contribute to the spread of misinformation. In this review, we examine misinformation sharing using a framework of human-algorithm interaction. We build upon recent arguments [12–14] that humans and algorithms need to be studied together as mutually interacting agents. Our framework predicts an important feedback loop which can help explain why people share misinformation: humans have pervasive attentional biases, such as those toward moral and emotional information; algorithms amplify moral and emotional information because it increases engagement on social media platforms, and misinformation exploits moral and emotional content in order to spread more widely via human-algorithm interaction. Understanding the spread of misinformation as a problem of human-algorithm interaction suggests several new interventions that target both human psychology and algorithm design.

Attention biases and algorithmic amplification

Humans do not process all the information in their environment. Rather, our attention acts as a sensory filter that processes certain information over other information [15]. In fact, humans have several attention biases that predict what types of information they are most likely to attend to, process and learn from [16,17]. For instance, humans disproportionately attend to moral [18–20] and emotional [16,21–23] information in the environment because such biases are functionally adaptive. Biases toward moral information aid in the regulation of social norms, including cheater detection [24–27] and penalizing norm violators to maintain cooperation [18,28–30]. Biases toward emotional information aid in the detection of environmental and social threats, which improves survival and group cohesion [31–34].

In online contexts, these generalized attentional biases emerge just as they do in offline contexts. Recent observational and experimental work has demonstrated that social media messages containing moral and emotional language spread widely online [10,35–44], and the ability of these messages to capture our attention helps explain why they spread [35,39].

Attention capture is only one element that explains why moral and emotional content spreads online. Aside from human attention biases, content algorithms on social media act as an exogenous attention filter by curating the information we are exposed to [45,46]. Content algorithms are specifically designed to increase user engagement and amplify content that maintains user attention since a large portion of platform revenue is drawn from advertising profits [46]. As a direct consequence of this function, content algorithms amplify the very information we are biased to attend to, including moral and emotional information.

Recent observational and experimental work has demonstrated that content algorithms on major social media platforms systematically amplify moral and emotional information such as anger- and outrage-inducing information [47], polarizing information [48], and extreme political information [49]. These studies support findings from leaked company documents released by whistleblowers that algorithms on platforms like Facebook systematically amplify anger-evoking content [50,51].

In summary, the interaction of human attention biases and algorithmic amplification on social media produces an information ecosystem that privileges moral and emotional information. In such an ecosystem, misinformation that exploits the amplification of moral and emotional content can spread further than misinformation that does not.

Misinformation uses moral and emotional information to spread

Our central argument is that the spread of misinformation online is driven by the interaction of human attention biases toward and algorithmic amplification of moral and emotional content. This argument is supported by our forthcoming work which found that misinformation exploits users' impulses to share moral outrage [52]. A combination of anger and disgust, moral outrage is typically experienced as a response to a perceived transgression against a moral norm and is one of the 'moral emotions' [53–57]. During contentious political events, as much as 20 % of politically engaged Twitter users' timelines can contain outrage [38]. Across 6 observational studies using data from Twitter and Facebook that spans two time periods and uses two operationalizations of misinformation, we found that misinformation evokes more moral outrage than trustworthy news (see Figure 1) [52]. We also found that news links that evoked higher levels of outrage were shared more on Facebook and Twitter. We confirmed this finding in a behavioral study: the odds of a participant being more willing to share a headline that evoked a high level of moral outrage were 2x the odds when outrage evocation was low. And while we observed this effect of outrage on sharing for both misinformation and

trustworthy news, our findings suggest that misinformation may be shared more frequently than trustworthy news because of its capacity to evoke higher volumes of outrage. In our observational Facebook Studies, we also found that outrage-evoking misinformation was more likely to be shared without being read first. Taken together, these findings suggest that outrage-evoking misinformation can be especially pernicious: it exploits biases in human attention and algorithm amplification, while also being more likely to bypass our ability to discern the accuracy of the news [52].

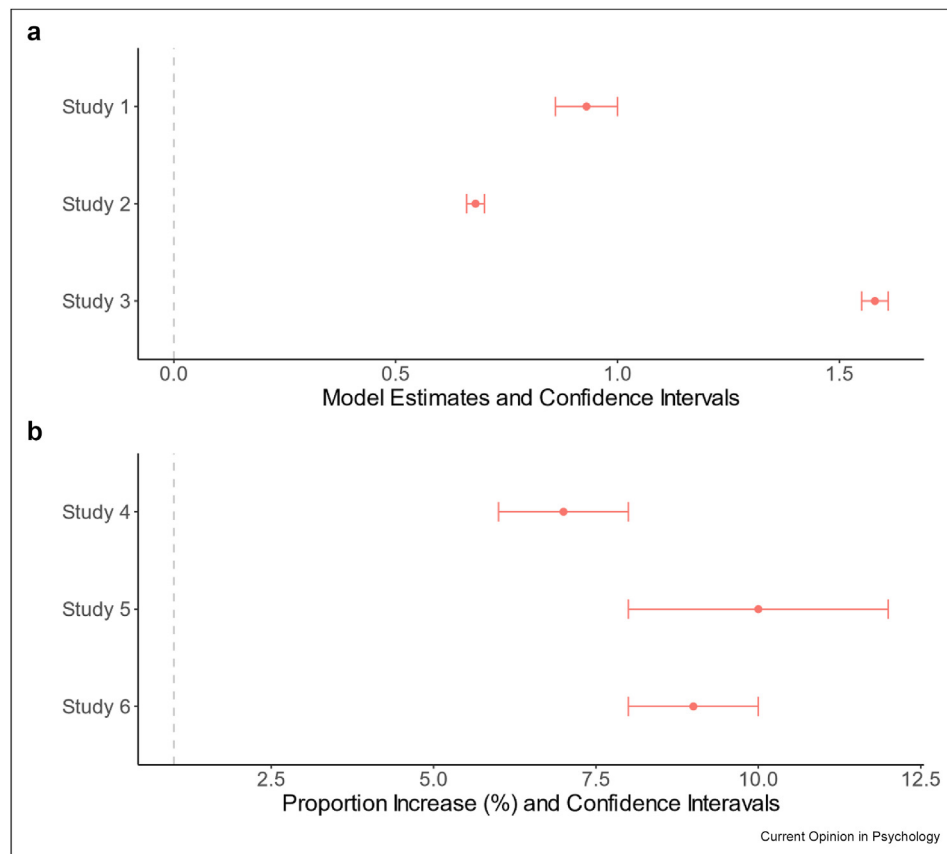
More generally, other recent studies have found that misinformation frequently contains emotionally-evocative content [58] and that being emotional – especially being angry – predicts susceptibility to misinformation and willingness to share it [59–61]. Although the spread of misinformation is multiply determined, preliminary work in the past five years strongly supports the idea that human attention biases to moral and emotional information interact with algorithmic amplification and play a role in the spread of online misinformation.

Intervening on human-algorithm interactions to combat misinformation

The perspective presented here suggests that focusing on the human psychology of misinformation sharing (e.g. attention to accuracy, partisan bias, attention to moral and emotional information) is necessary but not sufficient to combat its spread. To combat misinformation holistically, we suggest a dual-pronged approach that uses person-centered interventions *combined* with design-centered interventions [14,35].

Several person-centered interventions have already trained people to notice how misinformation exploits our attention to moral and emotional information [11,62,63]. For example, an intervention called *inoculation* increases resilience to misinformation by preemptively educating individuals about its perils [62,63]. However, recent work suggests that inoculations involving general warnings about the use of emotional language are not as effective as combining such inoculation with accuracy prompts [63]. Focusing peoples' attention to accuracy has been shown to improve their performance at identifying misinformation [11]. Yet, recent evidence suggests that in certain circumstances misinformation is spread in a habitized way that resists appeals to accuracy [52,64]. Importantly, when misinformation uses moral and emotional language it may cause people to share more habitually [54; Studies 1–3]. Additionally, it may make social motivations more salient, potentially at the expense of accuracy concerns [52]. These limitations of two well-studied person-centered interventions suggest that they may be insufficient for tackling the problem of misinformation sharing on their own.

Figure 1



Misinformation evokes more outrage than trustworthy news on Facebook **(a)** and Twitter **(b)**. *Note.* **(a)** Estimates and associated confidence intervals from models regressing outrage evocation on news type (misinformation versus trustworthy) using $N = 340,411$ links posted to Facebook shortly after the 2016 election (Study 1) and around the 2020 election (Studies 2 and 3). Estimates are for misinformation links relative to trustworthy links and all are significant (confidence intervals do not overlap zero): misinformation is more strongly associated with outrage than trustworthy news on Facebook. **(b)** The average increase in the proportion of outraged responses to tweets ($N = 61,922$) when they link to misinformation versus trustworthy news in 2016 (Study 4) and 2020 (Studies 5 and 6). Each increase is significant; on average linking to misinformation was associated with a nearly 10 % increase in the proportion of responses containing outrage. These results are from [52].

We suggest that these interventions should be combined with design-centered solutions that specifically counter the algorithmic amplification of moral and emotional content. Recent work supports the importance of modifying algorithm design in general. Across platforms as diverse as Twitter [65], Youtube [66], and Amazon [67], content algorithms have been observed to increase the prominence of misinformation in users' feeds and search results. Since the online news ecosystem is an interconnected network of different platforms, if even one platform deploys algorithms that amplify misinformation, then cross-platform traffic can cause users of all platforms to encounter algorithmically amplified misinformation [68].

We propose two specific design-centered solutions: *contextual down-ranking of moral outrage* and *bridging*. Contextual down-ranking of outrage [45] involves modifying algorithms to down-rank outrage content in contexts that are most likely to be associated with

misinformation (e.g. politics) [45,69]. This requires automated detection of outrage or other moral emotions at scale [38] combined with detection of high-risk contexts of information sharing. For example, algorithms could down-rank moral outrage content when outrage is detected alongside high rates of habitual information consumption (e.g., sharing without reading the article) [52]. An algorithm that down-ranks outrage in certain contexts could also be combined with a 'bridging' algorithm that up-ranks political information that is liked by both sides of the political spectrum, thus reducing the probability that the information has high partisan bias [70]. In general, the algorithmic solutions we present here share the important property of *flexibility*: down-ranking of information and notifications only in specific contexts based on dynamic measurement of moral, emotional, and political information [45].

One important consideration for the contextual down-ranking of outrage and bridging algorithms we

proposed is whether they threaten the business models of social media platforms by reducing engagement (making them unlikely to be adopted by companies). For example, if outrage content was down-ranked in political context, it might reduce engagement and advertising revenue. However, recent work suggests that outrage and political content are produced by a minority of individuals [71] and overrepresented by content algorithms in news feeds [49], leading the majority of users to actually prefer less intense political content in their feeds [72]. Thus, user experience may actually improve even though our attention is drawn to outrage content. Furthermore, interventions targeting human-algorithmic interaction can be implemented by organizations other than the social media companies. Civic society organizations, academics, and others have already developed or are developing interventions that can be implemented at the user level without the buy-in of companies. For instance, one group of researchers has developed a browser extension that allows users to minimize the ideological biases in their YouTube recommendations, without any cooperation from the platform [73].

A research agenda for the human-algorithm interaction framework

We have highlighted how content algorithms shape user behavior through a complex interaction with human psychology. This view complicates recent accounts that “user choice” is the dominant factor driving misinformation sharing, as opposed to algorithmic amplification [74]. The research we have highlighted suggests that neither human attentional biases nor algorithmic amplification on their own are sufficient to explain misinformation sharing - researchers need to explore their interaction and its emergent outcomes [45]. Behavioral scientists, computer scientists and developers should collaborate on ways to audit and de-bias content algorithms while also designing modifications to existing algorithms, allowing them to augment rather than exploit human attention biases that are not always adaptive in online ecosystems [14]. We offered two potential algorithmic modifications to combat misinformation (contextual down-ranking of moral outrage and bridging), but several others may prove useful in combating misinformation [75].

In order to examine misinformation through the lens of human-algorithm interactions, it is also important to understand algorithms as dynamic agents that change due to top-down platform decisions, as well as in response to changes in user activity. Future research may benefit from operationalizing content algorithms as dynamic artificial agents engaged in a back-and-forth learning with human users. Content algorithms learn from input from humans and networks, adjust their computations, and generate output accordingly. This

output then feeds back to the users and influences their behaviors [45]. Presently, much research ignores the complexity of these iterative interactions. However, existing tools like *breadboard* [76] and *Dallinger* [77] - which allow researchers to manipulate live participant networks - could potentially be adapted to encode agentic content algorithms and measure their interactions with participants over successive generations of social networks. This would allow for the measurement of emergent human-algorithm interactions, such as misinformation sharing, with high levels of ecological validity. Agent based simulation approaches - which manipulate human-like artificial agents, their interactions, and their environment [78] - could be used to simulate how small changes in the computations that content algorithms perform on human input affect the types of content that get amplified at the network-level.

This agenda can improve upon previous approaches by better capturing the complexity and emergent properties of human-algorithm interactions. However, such an agenda would also need to account for ‘top-down’ algorithmic changes - that is, changes driven by platform design compared to ‘bottom-up’ changes driven by human-algorithm interactions. The developers and owners of social media platforms have ultimate control over the rules that content algorithms follow. We know that deliberate algorithmic design choices ensured anger was up-weighted in the feeds of Facebook [51] and Twitter [47] users, but other top-down changes have also altered the recommendations of content algorithms. For instance, preliminary evidence suggests that Facebook’s deliberate decision to down-rank clickbait may have improved user experience and decreased engagement with clickbait on the platform [79]. The tools and methods that we have described could also be used to model top-down changes to the rules content algorithms use to make their recommendations and measure the resulting impact on users and networks.

Conclusion

While misinformation continues to proliferate at pace [8], the interactions between human users and content algorithms requires increased attention from psychologists. The research that we have reviewed here suggests that human attentional biases in favor of moral and emotional content play out online as much as, if not more than, they do offline. Since online content algorithms are designed to capture user attention, they end up amplifying the information we are biased to attend to, especially moral and emotional language. We have also discussed recent evidence that misinformation is particularly effective at evoking moral and emotional responses. The consequences of these strands of research are clear: misinformation that evokes moral emotions gains disproportionate traction online because of the interactions between human attentional biases

and algorithmic amplification, damaging public discourse and potentially skewing political outcomes. This suggests that interventions to combat misinformation sharing that are exclusively person- or design-centered miss the forest for the trees; misinformation sharing is an emergent byproduct of complex and dynamic interactions between human users and content algorithms that unfold online. Going forward, research should better operationalize the interactive role that content algorithms play in shaping online behaviors across time and at the individual and network levels.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

The authors thank participants of the misinformation preconference at the annual meeting of Society for Personality and Social Psychology (SPSP) for helpful feedback.

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- * of special interest
- ** of outstanding interest

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